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Smart if Small Economies?

Ireland's Strategy in Comparative Frame

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I. Introduction

Amidst the deepest and widest crisis to hit the capitalist world since the 1930s, government policy makers in many countries are induced to develop novel strategies aiming to promote new forms of industrial development and restore economic growth. In Ireland, the notion of the 'smart economy' has been advanced by government as a key concept to guide strategy for industrial restructuring, innovation and economic renewal.

Adopting a multidisciplinary theoretical framework, the primary aim of this paper is to interrogate the relatively new Irish policy concept of 'smart economy' by relating and comparing it to recent theories engaging with the characteristics of the changing industrial economy. Secondly, we carry out two empirical exercises that aim to better identify and understand the outcomes of past policies and the challenges that the new Smart Economy policies are facing.

The first of these two empirical exercises addresses the evolution of certain knowledge-intensive sectors of the economy. In performing this analysis we build on categorisations developed by Machlup (1962) and refined by Preston (2001) and we compare the evolution of output and employment of the 'core' and 'supporting' knowledge-intensive service industries. Broadly in line with Mjoset's (1992) example, we compare the evolution of Ireland's knowledge-intensive industries with the evolution of the same sectors in other small European countries, Japan and the United States.

The second empirical exercise focuses on innovation and creativity. In this case, we review Ireland's performance in trading knowledge intensive services and compare this performance with the one of a group of small European countries. Moreover, estimates regarding the number of patents issued are integrated in this analysis. In the conclusions, the comparison with other small European countries suggests that the Irish economy is more dependent on the presence of those industries that produce the tools favouring the creation and distribution of information, while the industries that produce information and knowledge in Ireland have experienced a slower growth over the last twenty years. Considering that the large majority of the former are owned by foreign capital, it is important to understand whether past policies have been successful in stimulating indigenous industrial innovation and creativity. The indications obtained from the second empirical exercise carried out in this analysis seem to signal that the level of indigenous innovation and creativity in Ireland is proportionally very low. Therefore, the findings from this analysis suggest that a national Smart Economy policy strategy should embrace a new paradigm which concentrates less on technical knowledge. At the same time, the new strategy should favour the development of those sectors and activities that produce the tools for the creation and diffusion of information and knowledge and concentrate more on developing indigenous, 'core' information industries.

II. The origins of Information and Knowledge as sources of Economic Growth

Many academics consider Daniel Bell as the most influential author of the mainstream theories of the information society (Preston, 2001). In his major thesis on the post-industrial or information society, Bell (1973) emphasises the rapidly expanding roles of relatively new specialist 'knowledge-intensive' services and occupations, especially certain specialist services and (computer-related) technical functions that had grown relatively rapidly in the USA from the late 1940s. These include high-level management and co-ordination or administrative functions as well as specialist scientific and technology based occupations in computing, electronic engineering, medical, economic and related research fields. Bell's thesis also relies heavily on the presumed role and implications of electronic computing systems in transforming both the character and role of knowledge. Nevertheless, although it also draws attention to the changing role of diverse knowledge specialisms or symbolic analysts, we may note that Bell (like Schumpeter before him) and most other conventional knowledge economy theories tend to neglect the expanding role of a certain sub-set of relevant services. These include the media and creative industries, not to mention the specific features and forms of specialist knowledge linked to (non-technical) communication occupations and functions which have continued to expand in late industrial societies. Such shortcomings could have been (at least partially) avoided if these modern and mainstream theories of the information society had paid more attention to aspects of Machlup's seminal contributions (e.g. 1962). We refer to his efforts to identify, describe and map one key aspect of the deepening division of labour since the turn of the twentieth century manifest in the relatively rapid growth of certain kinds of knowledge or information-intensive industries and occupations in the USA. Machlup and his followers (e.g. Porat), identified and analyzed the expanding role of a 'primary information sector', i.e. the growth of industries or services primarily engaged in the production, processing or distribution of various forms of knowledge or information, including those industries producing devices and systems supporting or enabling such activities. They also proceeded to identify a 'secondary' information sector, comprising those occupations or functions engaged in the production, processing, handling or distribution of various forms of information or knowledge in all other sectors of the economy (i.e. outside the primary information sector).¹

Surprisingly enough, the first significant appropriations of the information society idea by policy circles can be traced to the USA during the late 1970s and early 1980s. This arose when US foreign policy analysts concerned with international industrial and commercial strategies first identified the 'information society' idea as 'a strategic new element in the American global equation'. Indeed, these analysts defined the information society idea as one which well-matched strategic US industrial and trade interests, serving to 'amplify American ideas and values in a more forceful way than has ever been done before' (Dizard, 1982; cited in Preston, 2001: 73-74).

¹ This paper, in line with the practice of seminal theorists such as Daniel Bell, treats the terms 'information society', 'knowledge economy' and 'post-industrial society' as referring to broadly similar conceptual and theoretical claims or phenomena. However, I emphasise that these three can and must be distinguished from the descriptive concept of an 'information sector' (Preston, 2001).

We may also observe that this was the time of the Regan administration and the final demise of the era of 'we're all Keynesians now' as President Eisenhower of the USA defined it in the 1950s. More specifically, it witnessed the rise of neo-liberalism as the guiding ideology and practice in the USA's domestic and external policy affairs, especially as regards industrial and economic matters. Thus, it may be taken as hardly surprising to find that in its subsequent career in policy discourse, the information society idea has often been defined or presented as if hard-wired to the core tenets of neo-liberal doctrines. By the early 1980s, high-level advisers to the first Thatcher government in Britain were also moving towards somewhat similar concepts and industrial strategies centred on the business of 'making a business of information' (Preston, 2001).

At the EU level, interwoven moves and narratives around the neo-liberal vision of deepening globalisation of trade and investment, the liberalisation of telecommunications and other media related services, alongside the issues of 'competitiveness', were brought together and explicitly linked to the information society idea through a number of key policy documents published in the 1993-94 period. The most significant comprised a report from the then powerful Commissioner responsible for industrial and telecommunications affairs (EC, 1994a). Often referred to as 'the Bangemann Report', this policy document had major and lasting influences on the framing of subsequent policies for Information and Communication Technology (ICT) research and communication services, both at the EU level and in Ireland. Indeed, for some years following its publication, this report was repeatedly cited as a sort of 'bible' or master mantra by other Commission documents and by officials dealing with a very wide spectrum of industrial and social policy initiatives. For example, it was explicitly invoked as a framework for an influential 1994 document setting out a new industrial and policy strategy for the audiovisual sector in the context of the European Union's newly completed single market (Preston, 2001, 2003).

The Bangemann report's most obvious influence was to insert the term 'information society' as the key term in the vocabulary rather than Information Technologies (IT) or ICT in the EU's research, industrial and communications policy domains. This semantic shift was meant to reflect a greater emphasis on key applications services as well as demand-side rather than technology-centred approaches to research and industrial innovation. It also reflected a more (neo-)liberal view of the role of market forces and competition in the allocation of economic resources, including those related to telecommunications and to the selection and direction of new technological developments.

At the same time, not least for those associated with 'the Social Europe' agenda, the shift in vocabulary was taken or meant to imply something rather different. It was deemed to imply that there was, or may be, something special about 'Europe's Way to the information society' which reflected the traditional orientation towards a relatively robust welfare state and social democratic conceptions of citizenship rights in most if not all EU member states compared to the USA (EC, 1996a, 1996b). Thus, for the past decade or more, policy elites in Ireland, the EU and indeed worldwide, have appropriated the 'information society' or 'knowledge economy' as a meta-concept (or

master-myth) in framing discourses on the distinctive features and direction of contemporary social and economic developments.

Despite the semantic shifts inaugurated by the Bangemann report, however, these appear to have had little impact on subsequent policy initiatives, for example in the core R&D policy arena. Despite the substitution of information 'society' for 'technology', the vast majority of the ICT-related funding under the EC's Fifth Framework (FP5) research programme (1998-2002) continued to be allocated to the scientific and technical activities concerned with the further development of new technologies. The shift towards societal applications was more pronounced in the vocabulary framing the various research action lines than in the substance of the work undertaken (Preston, 2003; 2005).

We should also observe here that, since the year 2000, there has been a distinct marginalisation of the term information society in favour of the 'knowledge-based economy' in the framing of the EU's policy discourses and practices related to research and overall industrial and economic development goals. These semantic shifts have been directly influenced by the proceedings of the European Council meeting held in Lisbon in 2000 which expressed the ambitious aim of making Europe 'the most dynamic knowledge-based economy in the world capable of sustainable economic growth with more and better jobs and greater social cohesion' (EC, 2002a: 7). This goal has been linked to an expanding emphasis on the role of R&D for European competitiveness, especially at the European Council's Barcelona meeting in 2002 where the Heads of State and Government committed themselves to investing 3% of GDP in R&D by 2010 (Preston, 2003).

III. Smart Economy Policy Objectives and the Role of Knowledge and Information Services

The Irish government's first 'smart economy' document was published in December 2008. Essentially, this comprises, as the sub-title suggests, a '*Framework for Sustainable Economic Renewal*' in the face of the crisis that unfolded so dramatically earlier that year. The document is thus concerned with setting out 'the Government's vision for the next phase of Ireland's economic development' which will proceed by 'building on *the significant strengths of the economy*, which provide strong foundations for future *export-led economic growth*'. In this light, the report is presented as one which 'sets out an ambitious set of actions to develop Ireland's Smart Economy' (Ireland, 2008: 7).

Later in this document, the government insists that its 'framework consists of a set of interlocking elements each of which is reflected in a series of Action Points', and the five action areas identified also serve to 'demonstrate the specific measures which are being taken as a matter of urgency' (Ireland, 2008 : 9). These five key 'Action Areas of the Framework' are (Ireland, 2008: 9): (1) meeting the short-term challenge – securing the enterprise economy and restoring competitiveness; (2) building the ideas economy – creating 'the Innovation Island'; (3) enhancing the environment and securing energy supplies; (4) investing in critical infrastructure; and (5) providing efficient and effective public services and smart regulation.

This document also suggests that the government's 'Smart Economy' framework 'combines the successful elements of the enterprise economy and the innovation or 'ideas' economy while promoting a high-quality environment, improving energy security and promoting social cohesion' (Ireland, 2008: 7). It goes on to suggest that there are two key characteristics of the proposed strategy: 'A key feature of this approach is building the innovation or 'ideas' component of the economy through the utilisation of human capital – the knowledge, skills and creativity of people - and *its ability and effectiveness in translating ideas into valuable processes, products and services*. A second important aspect is the greening of the economy and the development of green enterprise (Ireland, 2008: 7). Furthermore, it is proposed that Smart Economy policies should aim at making Ireland "an innovation and commercialisation hub in Europe – a country that combines the features of an attractive home for innovative R&D-intensive multinationals while also being a highly-attractive incubation environment for the best entrepreneurs in Europe and beyond" (Ireland, 2008: 8).

The second substantive government document describing Smart Economy policies is the *'Science, Technology and Innovation – Delivering the Smart Economy'* published by the Department of Enterprise Trade and Employment in August, 2009. This report emphasises that 'economic renewal' must 'draw on our identified strengths' but it also 'requires the key policy priorities to be identified and the associated actions implemented (Ireland, 2009: 1). In this frame, the document then proceeds to identify and emphasise two 'key policy priorities': (1) "We require, firstly, to rediscover and embed the fundamental principle that a small, open economy must compete globally and be competitive globally. That reality must shape our responses to the daily pressures we face to sustain efficiency and drive higher value added output across the economy"; (2) 'Secondly, we must continue to invest in Ireland's research base as an important cornerstone underpinning our future competitiveness and capacity to innovate. A strong science base matched by an evolution in the capacity of our enterprise sector to create knowledge, to innovate, and to exploit new knowledge across global markets is critical to Ireland's future (Ireland, 2009: 1).

It is pertinent to observe that this document proceeds to re-emphasise many aspects of the 'vision' outlined in a prior (2006) *Strategy for Science Technology & Innovation* (SSTI) to the effect that 'Ireland, by 2013, will be internationally renowned for its research and be at the forefront in generating and using new knowledge for economic and social progress, within an innovation culture' (Ireland, 2009: 9). It suggests that this particular 'vision' of Ireland four years hence will be realised by 6 specific sets of actions (Ireland, 2009: 9): (1) Building a world class research system; (2) driving growth through research and innovation in enterprise; (3) capturing, protecting and commercialising ideas and know-how; (4) ensuring sound foundations in education; (5) focusing research in a number of sectors: Agriculture, Food, Health, Environment, Marine and Energy; and, (6) using research to drive economic and social return in Agriculture, Food, Health, Environment, Marine and Energy.

The policy document of August 2009 goes on to list a number of factors which help 'identify why advanced nations make significant investments in R&D (Ireland, 2009: 10): (1) Researcher talent attracts FDI companies and encourages them to stay; (2) Indigenous companies can use the know-how gained through technology transfer to

innovate and move up the value chain; (3) Patenting is a means of obtaining a return on the public investment; (4) Research in universities leads to the formation of new businesses by academics or graduates; (5) Students who have participated in advanced research can bring new skills and knowledge into existing enterprises. This second report also notes the government's action in 'setting up of an Innovation Taskforce to advance the development of Ireland as an international innovation hub and to assist in making the Smart Economy a reality' (Ireland, 2009: 11).

'In order to bring the SSTI through to fruition, the necessary graduates from third-level and increasingly nowadays, fourth-level, in science, engineering and technology will be required to come on stream. The SSTI focuses strongly on ensuring sustained development of human resources in science, technology and innovation to meet the needs of a rapidly growing knowledge society' (Ireland, 2009: 11).

In sum, the Irish government's 'Smart Economy' policies thus far, appear to be less novel than the new headline notion might imply. In many respects, they substantially represent a continuation of the policies carried out during the latter years of the so-called 'Celtic Tiger'. They highlight (rather familiar) policy actions attempting to foster economic growth by investing in technical knowledge and by favouring the development of those information industries' branches that rely on R&D. At the same time, we may observe a continuing neglect of other knowledge intensive services and industries.

The estimates presented in the next sections tend to emphasise, among other things, that there is an important gap to be bridged and filled in order to make Ireland an 'innovation island' capable of attracting foreign investments in knowledge intensive activities, given that, compared to other small European countries, the core information-intensive service industries, (those industries producing information and knowledge), seem to play a smaller role in the overall economy. Also, the data series and findings presented in the next sections indicate that policies of the recent past were indeed successful in helping certain sub-sectors become competitive in the international arena. At the same time, however, these policies also failed to produce much by way of export-led economic growth centred on enabling indigenous companies 'move up' the value chain and become providers of knowledge-intensive services or related service-based innovations.

IV. The Evolution of Ireland's Information Sector in a Comparative Frame

In order to compare the weight of different types of information industries in several European small economies, we build on and expand the primary information sector (PIS) category originally developed by Machlup's early work on knowledge-intensive activities. We apply the more refined and dis-aggregated PIS categories as advanced by Preston (2001), taking explicit care to avoid an overly techno-centric approach. Essentially, this means breaking down the various components of the PIS into a number of sub-sectors in order to undertake a comparative mapping of relevant shifts in industrial structures and performance over recent decades (as indicated in Table 1 below).

Applying this frame, we move on to examine and compare the evolution of output, employment, and trade performance trends as well other relevant indicators, distinguishing between 'core' and 'supporting' knowledge-intensive service industries in the selected countries.

Table 1: The Composition of the Primary Information Sector and sub-sectors

PIS Sectors	Components	KLEMS Code
PIS 1: Supply of ICT Devices, Systems and Networks	PIS 1b: New ICT Manufacturing Industries	
	PIS 1c&d: ICT Software, DP and Connectivity Services & Sel. Mail/Courier etc. connectivity services	
PIS 1b: New ICT Manufacturing Industries	Office, accounting & computing machinery	30
	Radio, TV, communication equipment	32
	Scientific instruments	331t3
PIS 1c&d: ICT Software, DP and Connectivity Services & Sel. Mail/Courier etc. connectivity services	Post and Telecommunications	64
	Computer and related activities	72
PIS 2: Core Information Industries	PIS 2a: Consumer Media and Cultural Industries	
	PIS 2b: Instrumental/ Producer IC Services	
	PIS 2c: Mixed' Producer-Consumer IC Services	
PIS 2a: Consumer Media and Cultural Industries	Print., publishing and recorded media	22
	Recreational, cultural and sporting activities	92
PIS 2b: Instrumental/ Producer IC Services	Activities auxiliary to financial intermediation	67
	Research and development	73
	Other business activities	74
	Activities of member organisations nec	91
PIS 2c: Mixed' Producer-Consumer IC Services	Education	M

Source: the authors. This division is developed based on the analysis presented in Preston, 2001.

IV.1 Data Series, Assumptions Tested and Data Sources

The primary objective of this first empirical exercise is to compare the evolution of the suppliers of ICT devices, systems and networks (i.e. the first major sub-sector of the Primary Information Sector, 'PIS1') to the evolution of the core information industries (i.e. the Primary Information Sector 2, or 'PIS2') as defined above. The assumption we are testing through this exercise is that too much emphasis has been placed on the PIS1's *direct contribution* as main transformation engine of economies, particularly during the years of the claimed New Economy (1995-2001), and too little on the role of the PIS2.

Our comparison of relative performance embraces some seven countries: Austria, Denmark, Finland, Ireland, Japan, Netherlands, Sweden and United States. In order to compare these trends, we use two types of indicators: output indices and an index of employment. These indices are estimated by calculating simple averages (as detailed below) from three series (value added at current, basic prices; value added

volume indices, and the number of employees) with respect to our selected seven countries.

The second objective of this exercise is to demonstrate that the statistical methodologies employed by national agencies have served to emphasize the importance of the PIS1 sector's direct contribution to economic change. The third objective of this exercise is to map and highlight Ireland's PIS's relative strengths and weaknesses, when this is compared to the trends for this same sector of other small and open European economies.

The dataset we employ for this exercise is drawn from the EU KLEMS database (online from <http://www.euklems.net/>). This database is the result of collaboration between eighteen research centres specialized in economic analysis from different European Countries. However, for the purpose of this exercise, we have used the series defined in the database as 'basic variables': (1) gross value added at current basic prices (in millions of local currency), (2) the number of employees, (3) prices' indices of gross value added, and (4) volume indices of gross value added. These 'basic variables' series are supplied by the different national statistical offices and are published by EU KLEMS database editors after minor manipulations.

Using the data extracted from the EU KLEMS database, first, we have calculated indices for the Primary Information Sectors and its various sub-sectors (1, 1b, 1c&d, 2, 2a, 2b, 2c, see table 1 above) for each of the seven countries composing our sample, by calculating average trends from the series of the industries composing these sectors. Second, we have also calculated trends 'average' PIS sectors and sub-sectors' series, from the series of all the countries in our sample. Third, as the period covered by this analysis is relatively long, i.e. from 1981 to 2005, for simplicity we have generated five intervals of five years. Therefore, the first interval has been calculated average trends using data from 1981 to 1985, the second from 1986 to 1990, the third from 1991 to 1995, the fourth from 1996 to 2000 and the fifth and last, from 2001 to 2005.

This procedure has been applied to all four 'basic variables' we have extracted from the EU KLEMS database. In summary terms, therefore, we have obtained and present our findings here related to:

- (1) A series showing the average growth of real value added of the industries that supply ICT devices, systems and networks for all the countries of our sample (table 2 in the appendices),
- (2) A series showing the average growth of real value added of the core information industries for all the countries of our sample (table 3 in the appendices),
- (3) Series indicating the average price and qualitative adjustments (i.e. the deflators) used by national statistical agencies to calculate the real value added growth of the industries that supply ICT devices, systems and networks (table 4 in the appendices) and of the core information industries (table 5 in the appendices).

IV.2 Output Growth Differentials between the PIS1 and the PIS2 Sub-Sectors

The indices composed from the series for value added real growth describe an important growth differential between the PIS1 and the PIS2 sectors. The average real output growth of the industries supplying ICTs has been much higher than the growth

of the core information industries over the entire period considered in this exercise. These averages indicate that already between 1981 and 1985, the PIS1 was growing much faster (i.e. by almost 9.5% a year) than the PIS2 (3%). Moreover, while the average real growth of the industries supplying ICT is 11.5% per year over the entire period (i.e. 1981-2005), the core information industries show a growth rate of only 3% (see tables 2 and 3 in the appendices).

Furthermore, the real output growth of the industries supplying ICT was almost 18% per year during the period when the 'New Economy' discourse was at its peak (i.e. the 1996-2000 period), while the same indicator for the core information industries shows an average growth of only about 4% per year. Also during the period that followed (i.e. between 2001 and 2005), characterized by a general economic slowdown, the real output growth of the PIS1 was much faster (i.e. over 8% per year, see table 2) than the real growth of the core information industries, which we estimate at below 2% (see table 3).

Interestingly, our estimates show that that real output growth of the industries supplying ICT in the U.S is mostly concentrated in the New Manufacturing Industry (i.e. the PIS1a, see table 1), while the real growth of the connectivity services, such as telecommunications, computer, post/courier services and related activities (i.e. the PIS1c and PIS1d sub-sectors) is relatively modest compared to the growth of the same sector in the other economies of the sample.

The Primary Information Sector in Ireland is found to be one of the most dynamic of the sample countries. According to our estimates, during the peak years of the so-called New Economy, the real growth of industries supplying ICT devices based in Ireland increased by almost 36% per year, while the real output of the core information industries grew by over 10%. Ireland's PIS is also one of the fastest growing of the sample over the entire period considered here: the PIS1 and PIS2 sectors experienced a 14% and a 6% average yearly growth, respectively. Interestingly, Ireland's Consumer Media and Cultural Industries (PIS2a) and the Mixed Producer-Consumer IC Services (PIS 2c) also seem to have experienced the fastest growth over the entire period when their trends are compared to the trends of the same sectors in the other countries of the sample (see tables 2 and 3 in the appendices).

So far, then, the analysis of real output growth based on our estimates seems to confirm the claims of those contributors to the 'New Economy' debates which celebrate the importance of the direct contribution of the ICT supplying industries to the sustained economic growth experienced by many western economies in the 1990s and in the beginning of the new millennium. However, the alternative estimates that we have calculated for relative shares based on current price index series turn up rather different trends.

IV.3 Output Growth Measures and the Importance of Deflators

The economic growth of industries, sectors or nations are most commonly analyzed using real (i.e. volume) output indices. These indices are obtained from current-price output series (i.e. based on observable market values) and calculated by using deflator series. Notably, the value of an industry's production expressed in current currency

can increase without an increase in the number of units produced. For example, this happens as a result of an increase in the price of inputs or as a consequence of the increase of the prices of other goods and services affecting the level of prices of different intermediate inputs. Moreover, thanks to various types of innovation, the characteristics of the same products or types of goods change over time.

Therefore, the rationale for using deflators is to homogenize goods and/or services which are different because produced at different times. The effect of deflators is twofold: first, they remove the influence of inflation, and second, they embody products' quality changes, when these changes are actually measurable.

Seemingly under the influence of the debate about an unfolding New Economy and the claimed mis-representation of the importance of the direct contribution from the ICT industries to the growth of the GDP, the United States' Bureau of Economic Analysis (BEA) in the 1990s adopted novel metrics. These comprised hedonic price measures (i.e. based on measurable characteristics) for computers and peripheral equipment, semiconductors, pre-packaged software, and digital switching equipment (Nordhaus 2002). Our analysis leads us to suggest that these changes and, more generally, the use of deflators, tend to emphasize the output growth of those industries producing goods, whose qualitatively changes are measureable, and to undermine the rate of output growth of other activities (notably, service activities) when these two types of output are compared or aggregated.

National statistical agencies generate deflators using different methodologies; therefore in order to discuss the consequences of applying deflators to ICT supplier industries and core information industries' output indices, we have created deflator series for the PIS1 and PIS2 sub-sectors for each country by following the same methodology described above and for the same intervals. The results of this exercise show that the growth rate of the average price used to estimate the real value added series of the industries supplying ICT devices (PIS1) is positive (4.64%) for the period 1981 to 1985, and negative for each subsequent period. On the other hand, the growth rate of the average price used to calculate real value added index series of the core information industries is positive and decreasing from the first to the last interval (see table 4 and 5 in the appendices).

It may be noted that a negative trend in the price series implies a 'quality-adjustment effect' that is more important than the 'inflation-adjustment effect'. This means that the value added volume index grows faster than the corresponding index built on the current-price value added series. Therefore, although the extent of the difference between the two series also depends on the base year chosen (1995 in this case), the application of deflators clearly affects the trends of the output of the industries supplying ICT devices and the output of the core information industries in opposite ways: it boosts the growth of the PIS1 real value added series while it reduces the growth of the PIS2 series.

We also observe that the difference in the trends designed by the deflators used to calculate real output growth trends for the PIS1 and PIS2 is also quite remarkable: for example, for the period 1996-2000, the average deflator series calculated to adjust the PIS1 current value added series of the countries considered in our sample decreases

by almost 3% per year, while the price series used to estimate the PIS2 real value added series shows an increase of 3% per year.

Among the deflator series estimated by the different national statistical agencies and used for this exercise, the trends described by the series used to estimate the real output of the industries supplying ICT devices and based in the U.S. are the most striking. These price series show that a negative adjustment of over 7% per year is applied to the current value output of the PIS1 sub-sector and an adjustment of almost 19% per year is applied to the current value added output series of the New ICT manufacturing industries sub-sector.

Therefore, the adoption of deflators clearly emphasizes the role played by increasing quality in the case of manufacturing industries' output. The positive trend of deflators used to calculate the real value of service industries' output, on the other hand, assume that the value of the latter increases mostly due to the increase of prices. However, as we have explained elsewhere, in the case of certain products and services, a growing price is likely to be indicative of increasing quality (Preston and Sparviero, 2009; Sparviero and Preston, 2010). This arises, for example, when the cost of supplying 'creative inputs', defined as original ideas, concepts, actions and inductive solutions to ill-defined problems, represents a large share of the total production costs of these 'creative' products and services. Therefore, we find interesting to compare the trends of real growth series with the trends designed by alternative measures.

IV.4 Assessing the Growth of GDP and Employment using Relative Shares

Since we observe certain limitations in using real output growth series for comparing the differences in the growth of Primary Information sub-sectors, we have also sought to compare these trends to estimates of relative shares that we have constructed from the more 'unrefined' current price value added series. Therefore, in order to assess the relative importance of the PIS and its sub-sectors, we have aggregated the value added of the industries composing these sectors and then calculated their respective shares of Gross Domestic Product (GDP). This enabled us to calculate averages for the same time intervals used before and, also, we calculated averages for the sectors from the series of all the countries in the sample (see tables 6 and 7 in the appendices). We also applied this procedure to the data related to the numbers of employees in order to obtain relative shares of employment (see tables 8 and 9).

It may be noted here that, the relative sizes of PIS and its sub-sectors during a given interval depend on the definitions of these sectors: i.e. the groups of industries that we have included under the different headings. Therefore, more interesting are the estimates of the changes to the relative sizes between periods, as these indicate if, and to what extent, the group of industries that we have classified here as core information industries and the industries supplying ICT devices, systems and networks, have grown relative to other industries, and how these industries have performed depending on the economies where they operate.

Therefore, we calculated the yearly changes in the relative size of current value added and employment of each sub-sector and for every country in the sample, and then, we calculated the average growth for our time intervals (see tables 10, 11, 12 and 13 in the appendices). By adopting this methodology we make some implicit assumptions:

we assume that, overall, qualitative changes and the effect of prices affect the different industries in symmetrical ways.

The resulting estimates show that, since the early 1980s, the PIS is constantly expanding in relative terms. The averages calculated across the countries of our sample show that in the beginning of the 1980s the total value added of the industries that supply ICT devices, systems and networks (PIS1) was roughly equivalent to 4% of the GDP and to a third of the value added of the activities composing the core information industries (12.5%). However, by the first years of the new millennium, the former had reached a relative output size of 6.8% of the GDP and almost half of the relative share of latter (16%, see tables 6 and 7 in the appendices).

Rather similar trends can be observed for our estimates of relative employment: the PIS1 employed 4.3% of the total between 1981 and 1985, while almost 16% of the active population of our sample countries, on average, was employed by PIS2 industries. By the period 2001-2005, the total labour employed in average PIS1 industries was equivalent to over 5% of the total, while almost 21% was employed by PIS2 industries (see tables 8 and 9 in the appendices).

If we turn our attention to the growth dynamics of the relative shares of value added, we notice that, the period where the PIS1 industries have grown faster than the other industries is, as expected, the interval 1996-2000. On the other hand, the average PIS2 shows the fastest growth during the second interval, i.e. the period from 1986 to 1990. Overall, if we calculate a total average of yearly growth for the entire period (i.e. from 1981 to 2005), the core information industries seems to have grown faster in relative terms (0.8%) than the industries that supply ICT devices, systems and networks (0.6%, see tables 10 and 11).

On the other hand, if we focus on the growth dynamics of the relative shares of employment, we observe that employment in core information industries is growing in every interval of our timeline and particularly in the interval 1991-1995 (i.e. the period that follows the fastest, value added growth). On the other hand, the growth of relative employment in the PIS1 peaked in the 1996-2000 period. On average and over the entire period, employment grows in relative terms faster in the core information-intensive service industries (1.2% per year) than in the industries that supply ICT devices, systems, networks (0.17%, see tables 12 and 13).

IV.5 Some Characteristics of Ireland's Primary Info Sector & its sub-sectors.

One of the objectives of this analysis was to identify some of the key characteristics of Ireland's primary Information Sector and its component sub-sectors. Our estimates show that, during the interval from 1996 to 2000, total employment in the new ICT supplying industries was equivalent to 7% of the total (see table 8 in the appendices), while the output of these industries equalled 9.6% of the total economy, the largest relative output for that period among all the countries in the sample (see table 6 in the appendices). However, both relative output and employment of these industries decreased between 2001 and 2005: the former by 4.4% per year, while the latter by almost 1.3% per year (see tables 10 and 12 in the appendices). In the last interval, our estimates show that the output of the industries that supply ICT was 8.8% of the GDP

and second only to Finland, while the relative employment was still some 6.75% of the total and the highest of the sample countries (see tables 6 and 8).

On the other hand, Ireland's core information sector is much less dynamic when it is compared to the same aggregate sector of the other countries in the sample: our estimates show that between 2001 and 2005, the output of the PIS2 was equivalent to 15.4% of the total, below the average of the sample countries (see table 7 in the appendices), while the level of employment is at 16% of the total, the lowest when compared to the other countries (see tables 9 in the appendices).

IV.6 Summary of the Key Trends Identified Here

In summary, the estimates we have calculated using the EU KLEMS database provide the following indications: the industries that supply ICT devices, systems and networks have experienced a very fast growth of production during a relatively short period (mostly, in the second half of the 1990s), while the core information-intensive services industries have experienced a slower, but steadier growth from the 1980s to 2006, the last year available for this exercise.

Moreover, we have observed that output volume indexes can lead to rather different sets of conclusions than those suggested by relative output shares built on current (basic) price series, as deflators boost the 'real' value of the PIS1's production to the extent that their average volume growth over the entire period considered here is faster than the average growth of the core information-intensive services industries. In contrast, we observe that when using current price values and relative shares, although the industries supplying ICT devices expanded faster during the mid 1990s, throughout the 25 years of our sample the core information industries seem to have grown more than the industries that supply ICT devices, systems and networks.

This differential in trends between these two major components of the Primary Information Sector is even more pronounced when we focus on the economic trends that have characterized the Irish economy: the production of the industries that supply ICT devices, systems and networks based in Ireland expanded even faster during the period from 1996 to 2000 than the output of the same industries based in the other countries selected here for comparison purposes. However, the slowdown in growth was also sharper in the following interval (i.e. from 2001 to 2005, see tables 2 and 10). Moreover, we also notice that, compared to the countries in our sample, the relative size of the industries using and producing information and knowledge (i.e. the PIS2) based in Ireland, is one of the largest in terms of production and one of the smallest in terms of employees (see tables 7 and 9). Also, in the core information industries based in Ireland have experienced a smaller relative growth within the timeframe used for this analysis.

V. The 'Creative Island's Strengths and Weaknesses

In this section, as indicated above, we will review a further series of indicators that will help advance our understanding of the extent to which Ireland's capacity to innovate has improved in the period since the 1980s and in particular, during the late 1990s period and its immediate aftermath. As in the previous exercise, we will seek to

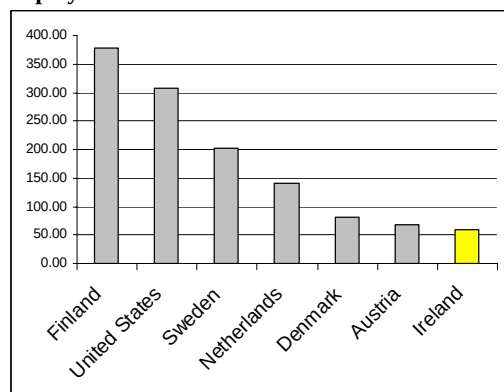
compare Ireland's performance to the performance of other small and open European economies.

First, in this section, we consider Ireland's capacity to innovate according to the more traditional notions of innovation by reviewing indicators for trends related to the production of patents. We then move on to embrace a broader, late-modern notion of innovation and apply this in a review Ireland's performance in trading key categories of 'creative' goods and services.

The indicators examined here to estimate the output of innovation efforts fostered by R&D are (1) the total number of patents granted by the European Patents Office (EPO, see table 14 in the appendices), (2) the total number of patents in the field of Information and Communication Technology granted by EPO and the United States Patents and Trademark Office (USPTO, see table 16), and (3) the total number of patents in the field of Biotechnologies granted by EPO and USPTO (see table 15). For comparison purposes, we standardise the number of patents issued to different stakeholders by using the number of total employees. The countries chosen for this exercise are Austria, Denmark, Finland, Ireland, the Netherlands, Sweden and the United States, while the data refer to the years 1985, 1990, 2000, 2005 and 2006.

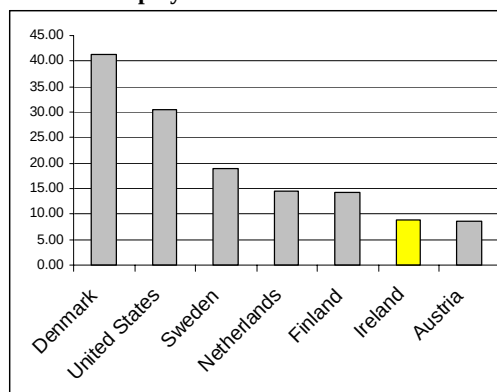
These indicators show that number of patents granted by the EPO to companies and individuals based in Ireland (and therefore, also the number of patents per thousand of employees) has surprisingly declined between 1996 and 2000. However, in general, our estimates indicate that, with the exception of this period, innovation performance related to R&D has steadily increased between 1985 and 2005. Nonetheless, in 2006, the Irish economy has produced the equivalent of less than 5 patents per thousand of employees. The only small country in the sample that has produced a similar, relative number of patents is Austria, while the next small European economy from the bottom of this particular classification is Denmark with over 17 patents per thousand of employees. Moreover, the small European economy that in 2006 was the most prolific in producing patents was Finland, as the number of patents issued by the EPO in that year to Finnish firms and individuals was equivalent to 30 per thousand of employees in the country (see table 14 in the appendices).

Figure 1: Number of ICT related Patents awarded by the EPO and USPTO in 2006, per million of employees



Source: Table 16 in the appendices.

Figure 2: Number of Biotechnology related Patents awarded by the EPO and USPTO in 2006, per million of employees



Source: Table 15 in the appendices.

In this section, we also focus on patents related to innovations in the Biotechnology and ICT sectors as granted by EPO and the USPTO. As the number of patents awarded for these types of innovations is only a sub-section of the total number of patents and therefore much smaller, we standardise these figures by using series of million of employees as denominators. These indicators show that the number of patents produced by firms and individuals based in Ireland is the lowest among the country of our sample, for both industries and for all the available years of these series. In 2006, the total patents related to the biotechnologies granted to Irish companies and individuals were equivalent to almost 9 per million of employees. While Austria reached approximately the same level, companies and individuals based in Finland were awarded the equivalent of over 14 patents per million of employees. All the other countries in the sample have produced a higher, relative number of patents than the industries and individuals based in these countries, with Denmark leading this particular standing with over 41 patents per thousand of employees (see table 15 in the appendices).

These estimates suggest that innovators based in Ireland and Austria occupy the bottom places of the standing built from the data of ICT related patents for the year 2006, respectively, with (almost) 60 and 67 patents per million of employees. Third from the bottom of this standing are industries and individuals based in Denmark, where the number of patents issued in 2006 is equivalent to over 80 per million of employees. Innovators in the ICTs based in Finland were by far the most prolific as they were awarded a number of patents equivalent to more than 378 per million of employees (see table 16 in the appendices).

VI. The Relative Performance of Knowledge Intensive industries

The purpose of this section is to analyse the trading performance of a selection of knowledge intensive services based in Ireland and to compare it to the other countries of the sample. This selection of service activities include Telecommunication services (table 17 in the appendices), Financial Services (table 18), Computer and Information Services (table 19), Personal, Cultural and Recreational Services (table 20), Royalties and licence fees (table 21), and Other Business Services (table 22). Unfortunately, most of these are rather broad or 'major' categories and so more fine-grained trends within these categories cannot be established as the requisite detailed and standardised (comparable) data for the relevant sub-sectors are simply not available in most cases.

The available data suggest that, compared to the other countries in the sample, the relative strengths of the Irish economy reside in the positive trend balance in the 'Computer and Information Services' and 'Financial Services' sectors (see tables 21 and 22). The data for the last year available (i.e. 2006) shows that the contribution to the Irish GDP from the Computer and Information Services industries based in Ireland was over 9%, while the net surplus from the activities classified as Financial services was equivalent almost 1.5% of the GDP. In contrast, none of the other countries in the sample, indicated any similar significant contribution to GDP (i.e. over 1% of the GDP, for example) from the trade balance generated by the computer and information services and financial services.

However, overall, it seems that during the same year (2006) the contribution from the trade generated by the primary information sector's service activities (at least the ones discussed in this section) was negative. The contribution from royalties and licence fees was negative and equivalent to almost 9% of the GDP; also negative (-6.3%) was the contribution from the industries classified under the heading of 'Other Business Services', which includes technical and legal consultancy services, as well as R&D (see tables 24 and 26 in the appendices). In particular, the contribution from this latter was equivalent to almost 2% (see table 22). On the contrary, the GDP of most the other countries in the sample benefited from the trade of trademarks, copyrights, patents, processes, techniques, designs, manufacturing rights and franchises (i.e. royalties and licence fees); in relative terms, this contribution range from Denmark's 0.13% to Sweden's 0.59%. On the other hand, the GDP of Austria and Finland was reduced by the trading of royalties and licence fees by 0.24% and 0.11% respectively. Moreover, still in 2006, the contribution to the GDP of the activities included in the 'Other Business Services' sector was positive for all the other countries in the sample. The extent of this contribution ranged from Denmark's 0.07%, to Finland's 2.34% (see table 21).

On the positive side, this particular empirical analysis suggests that the Irish economy manifests a relative strength in delivering certain information services compared to other countries. However, at the same time the estimates also point to a significant weakness with respect to the competitiveness of certain information or knowledge-intensive services, especially those that may be designated as 'creative'.

In particular, such findings seem to underline the extent to which, despite the much-vaunted Celtic Tiger' phenomenon, the Irish economy seems to remain highly dependent on innovations and knowledge resources (IPRs) that are owned and controlled by overseas entities. For example, compared to other countries examined, its expenditures on foreign R&D outputs is relatively significant, as is the extent of resources spent on trademarks, copyrights, patents, processes, techniques, designs, manufacturing rights and franchises produced outside of its territory.

VII. Conclusion

In the analysis reported in this paper we have sought to map and compare the relative importance of information and knowledge intensive sector in Ireland and several other economies. To this end, we have drawn on a series of basic economic indicators in combination with applying a distinct a classification of activities first developed by Machlup (1962) and then refined by Preston (2001).

The most important result that we have drawn from this initial empirical exercise is that, compared to the other countries in the sample, the Irish economy has experienced a more important relative growth of the industries supplying ICT devices, systems and networks (i.e. the PIS1 sub-sector of the Primary Information Sector) than the core information industries (i.e. the PIS2 sub-sector). We also observed that the growth of this sector in Ireland was particularly pronounced in the second half of the 1990s.

The analysis of this first set of indicators needs to be integrated with the results obtained from a second empirical exercise carried out in this paper where we have used several additional indicators to estimate the level of innovation and creativity of the Irish economy and to compare it to the performance of other small European countries. This analysis shows that, although Ireland's 'Computer and Information Services' and 'Financial Services' performed well in recent years, the benefits realised by the surplus apparently generated by these sectors are cancelled by the flow of money spent on trademarks, copyrights, patents, processes, techniques, designs, manufacturing rights and franchises registered abroad. Moreover, the Irish economy also presents a negative trade balance in other important knowledge intensive industries such as R&D and different types of consultancy services (included in the 'Other Businesses' categories).

In combination, these empirical exercises seem to suggest that, compared to other small and open European countries, the Irish economy suffers from a low level of indigenous innovation and creativity. This result is emphasized by the analysis of data regarding the relative production of patents, as the indicators we have built for this international comparison shows that Irish economy performed poorly.

As a result, these indicators taken together show that Ireland has seemingly lost an opportunity during the period of the Celtic Tiger to build strong, indigenous and innovative knowledge intensive industries' sector. The findings serve to underline the very considerable barriers and challenges facing the government if it is to realise its current strategic objectives -- to build the smart economy and an 'innovation island'—especially in a period marked by deep recession, low growth prospects and restricted resources. The implications of such findings for any viable 'smart economy' strategy seem quite clear, especially in terms of the requisite industrial, innovation, research and education policies. A thoroughgoing, veritable 'paradigm shift' in policy thinking and practices is urgently required in all these fields—tendencies to rely on a mere continuation of past policies (as evident in current policy documents) are like to result in dismal failure. The requisite paradigm shift, we suggest, should concentrate on a much more balanced distribution of resources fostering, on the one hand, technical innovation based on R&D, and, on the other hand, non technical innovations that are the vital to the development of knowledge intensive services – and fully recognise the diversity of services functions and knowledge fields which can support and deliver successful innovation performance in the 21st century economy. It must move beyond the old habits of fetishising technology whilst neglecting the array of other knowledge forms and fields that can support and enable innovation in any 'smart' economy worthy of the name.

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Appendices

Table 2: Value Added, Average Yearly Growth, PIS1

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Average 1981-05	Notes
1	Supply of ICT Devices, Systems and Networks						
<i>Average</i>	9.49%	11.84%	9.90%	17.99%	8.18%	11.48%	
Austria	6.68%	7.79%	4.87%	18.46%	6.30%	8.82%	
Denmark	13.10%	18.14%	7.57%	15.24%	5.52%	11.91%	
Finland	13.51%	12.55%	12.17%	13.38%	21.35%	14.59%	
Ireland	12.67%	7.71%	10.76%	35.91%	3.63%	14.13%	
Japan	8.28%	24.00%	7.31%	8.47%	6.61%	10.93%	
Netherlands	2.76%	9.83%	12.37%	14.29%	-0.37%	7.78%	
Sweden	3.69%	3.26%	9.94%	14.03%	12.11%	8.61%	
United States	15.20%	11.43%	14.23%	24.12%	10.33%	15.06%	
1b	New ICT Manufacturing Industries						
<i>Average</i>	13.59%	16.45%	14.47%	20.93%	10.82%	15.25%	
Austria	7.33%	7.03%	2.58%	26.08%	6.17%	9.84%	a
Denmark	16.66%	24.39%	9.97%	15.31%	2.87%	13.84%	a
Finland	17.81%	18.55%	22.16%	10.78%	36.47%	21.15%	a
Ireland	25.56%	13.39%	15.94%	33.68%	4.92%	18.70%	
Japan	14.18%	42.65%	9.45%	10.42%	9.73%	17.29%	
Netherlands	1.75%	9.95%	18.87%	9.84%	-5.69%	6.94%	a
Sweden	4.16%	1.22%	15.41%	20.53%	18.18%	11.90%	a
United States	21.29%	14.43%	21.38%	40.82%	13.93%	22.37%	
1c&d	ICT Software, DP, Sel. Mail/Courier etc. and connectivity services						
<i>Average</i>	5.38%	7.23%	5.33%	15.04%	5.55%	7.71%	
Austria	6.03%	8.56%	7.16%	10.83%	6.43%	7.80%	
Denmark	9.54%	11.89%	5.17%	15.16%	8.16%	9.99%	
Finland	9.20%	6.55%	2.18%	15.98%	6.23%	8.03%	
Ireland	-0.23%	2.02%	5.58%	38.15%	2.33%	9.57%	
Japan	2.38%	5.36%	5.17%	6.52%	3.49%	4.58%	
Netherlands	3.77%	9.72%	5.87%	18.73%	4.95%	8.61%	
Sweden	3.22%	5.30%	4.47%	7.54%	6.04%	5.31%	
United States	9.12%	8.43%	7.08%	7.42%	6.74%	7.76%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 3: Value Added, Average Yearly Growth, PIS2

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Average 1981-05	Notes
2	Core PIS information Industries						
<i>Average</i>	3.04%	3.96%	1.79%	4.11%	1.85%	2.95%	
Austria	0.99%	2.98%	2.39%	4.25%	2.21%	2.56%	b
Denmark	2.10%	7.17%	1.22%	2.82%	0.54%	2.77%	
Finland	6.53%	3.96%	0.71%	5.45%	-0.15%	3.30%	
Ireland	2.25%	4.82%	4.57%	10.28%	7.04%	5.79%	
Japan	5.41%	3.36%	0.35%	0.93%	1.01%	2.21%	
Netherlands	0.74%	3.28%	1.37%	2.55%	0.31%	1.65%	c, f
Sweden	1.90%	1.76%	1.49%	2.52%	2.35%	2.00%	
United States	4.41%	4.38%	2.24%	4.05%	1.48%	3.31%	
2a	Consumer Media and Cultural Industries						
<i>Average</i>	2.35%	4.01%	1.22%	4.32%	1.33%	2.65%	
Austria	0.72%	6.29%	0.18%	5.38%	1.06%	2.73%	b
Denmark	0.74%	0.49%	0.86%	1.46%	0.34%	0.78%	
Finland	4.64%	2.84%	-0.90%	3.46%	-0.64%	1.88%	
Ireland	2.53%	5.53%	5.93%	15.09%	9.52%	7.72%	
Japan	4.15%	6.54%	-3.23%	-0.79%	0.65%	1.47%	
Netherlands	0.31%	6.09%	1.50%	3.44%	0.01%	2.27%	c
Sweden	1.27%	1.28%	3.17%	1.43%	-0.28%	1.37%	
United States	4.45%	3.05%	2.26%	5.07%	-0.06%	2.95%	
2b	Instrumental/ Producer IC Services						
<i>Average</i>	4.84%	5.64%	3.32%	6.75%	3.06%	4.72%	
Austria	0.61%	2.44%	6.59%	5.95%	4.68%	4.06%	
Denmark	4.50%	21.21%	0.58%	4.68%	0.98%	6.39%	
Finland	13.86%	7.13%	3.13%	11.27%	-0.45%	6.99%	
Ireland	2.70%	1.22%	3.86%	15.13%	9.76%	6.54%	d
Japan	7.91%	-0.57%	5.68%	3.23%	1.23%	3.50%	e
Netherlands	-0.19%	4.28%	2.50%	4.23%	0.51%	2.27%	f
Sweden	2.72%	2.79%	1.58%	5.08%	5.25%	3.48%	
United States	6.64%	6.65%	2.60%	4.44%	2.50%	4.57%	
2c	Mixed' Producer-Consumer IC Services						
<i>Average</i>	1.93%	2.23%	0.84%	1.25%	1.16%	1.48%	
Austria	1.65%	0.19%	0.40%	1.41%	0.88%	0.91%	
Denmark	1.08%	-0.20%	2.21%	2.32%	0.31%	1.14%	
Finland	1.08%	1.92%	-0.11%	1.63%	0.64%	1.03%	
Ireland	1.51%	7.72%	3.91%	0.61%	1.84%	3.12%	
Japan	4.18%	4.11%	-1.41%	0.34%	1.13%	1.67%	
Netherlands	2.09%	-0.52%	0.10%	0.00%	0.42%	0.42%	
Sweden	1.71%	1.20%	-0.26%	1.03%	2.09%	1.15%	
United States	2.14%	3.45%	1.86%	2.64%	2.00%	2.42%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 4: PIS1 Deflators (base 1995)

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Notes
1	Supply of ICT Devices, Systems and Networks					
<i>Average</i>	4.64%	-0.16%	-2.12%	-2.93%	-3.48%	
Austria	3.10%	1.82%	1.93%	-1.64%	-3.13%	a
Denmark	2.14%	-4.50%	-5.47%	-3.59%	-3.10%	a
Finland	9.08%	5.01%	-3.71%	-3.38%	-0.66%	a
Ireland	10.44%	1.41%	2.52%	2.10%	-2.46%	
Japan	0.73%	-4.96%	-3.09%	-5.40%	-5.97%	g
Netherlands	2.21%	-2.21%	-4.89%	-1.37%	-1.45%	
Sweden	7.78%	5.60%	0.20%	-2.86%	-3.86%	a
United States	1.65%	-3.45%	-4.43%	-7.30%	-7.18%	
1b	New ICT Manufacturing Industries					
<i>Average</i>	2.54%	-2.96%	-5.89%	-5.95%	-6.11%	
Austria	0.29%	0.43%	0.96%	-1.26%	-4.86%	a
Denmark	-3.66%	-8.87%	-8.65%	-3.39%	-3.81%	a
Finland	11.34%	5.11%	-8.33%	-7.50%	-1.31%	a
Ireland	7.61%	-0.97%	1.73%	2.43%	-6.86%	
Japan	-0.07%	-11.40%	-6.84%	-7.86%	-9.35%	g
Netherlands	2.23%	-5.12%	-10.85%	-1.05%	-3.60%	
Sweden	8.11%	5.06%	-5.12%	-10.20%	-6.42%	a
United States	-5.50%	-7.91%	-9.99%	-18.78%	-12.64%	
1c&d	ICT Software, DP, Sel. Mail/Courier etc. and connectivity services					
<i>Average</i>	6.73%	2.64%	1.65%	0.09%	-0.85%	
Austria	5.91%	3.22%	2.90%	-2.01%	-1.39%	
Denmark	7.93%	-0.12%	-2.28%	-3.79%	-2.39%	
Finland	6.81%	4.92%	0.91%	0.74%	-0.01%	
Ireland	13.27%	3.80%	3.32%	1.77%	1.93%	
Japan	1.53%	1.47%	0.66%	-2.95%	-2.59%	
Netherlands	2.18%	0.70%	1.06%	-1.69%	0.70%	
Sweden	7.46%	6.15%	5.52%	4.47%	-1.30%	
United States	8.79%	1.00%	1.12%	4.18%	-1.73%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 5: PIS2 Deflators (base 1995)

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Notes
2	Core PIS information Industries					
<i>Average</i>	6.94%	5.25%	3.71%	3.06%	2.78%	
Austria	5.50%	3.90%	4.63%	1.51%	1.89%	b
Denmark	7.89%	6.63%	2.63%	2.94%	2.35%	
Finland	9.99%	7.64%	3.35%	1.91%	4.27%	
Ireland	11.76%	6.00%	5.18%	5.67%	5.00%	
Japan	3.13%	3.49%	1.90%	0.58%	-0.85%	f
Netherlands	1.02%	1.76%	3.19%	3.97%	3.59%	
Sweden	8.80%	8.32%	5.06%	4.88%	2.75%	
United States	7.44%	4.22%	3.73%	2.98%	3.23%	f
2a	Consumer Media and Cultural Industries					
<i>Average</i>	6.98%	5.27%	3.47%	2.57%	1.94%	
Austria	4.78%	4.11%	5.50%	0.66%	2.06%	b
Denmark	8.74%	7.44%	2.83%	2.76%	0.33%	
Finland	8.17%	7.14%	1.89%	0.79%	3.89%	
Ireland	11.42%	5.79%	4.63%	5.39%	0.87%	
Japan	4.00%	2.99%	2.51%	0.76%	-0.64%	
Netherlands	2.13%	1.59%	2.36%	2.95%	2.62%	
Sweden	9.83%	8.39%	4.36%	4.66%	3.02%	
United States	6.81%	4.68%	3.71%	2.58%	3.37%	
2b	Instrumental/ Producer IC Services					
<i>Average</i>	7.28%	5.47%	4.25%	3.76%	2.63%	
Austria	6.05%	3.33%	4.18%	2.44%	1.96%	
Denmark	7.17%	6.86%	2.76%	3.69%	3.32%	
Finland	10.03%	7.62%	5.33%	2.70%	4.90%	
Ireland	11.69%	7.04%	5.88%	6.88%	3.25%	
Japan	3.59%	4.30%	1.20%	0.50%	-0.83%	f
Netherlands	2.49%	1.75%	3.19%	3.83%	3.02%	
Sweden	9.64%	8.52%	7.62%	6.19%	1.87%	
United States	7.58%	4.33%	3.86%	3.84%	3.53%	f
2c	Mixed' Producer-Consumer IC Services					
<i>Average</i>	6.56%	5.00%	3.40%	2.84%	3.77%	
Austria	5.67%	4.26%	4.20%	1.42%	1.64%	
Denmark	7.75%	5.61%	2.31%	2.36%	3.39%	
Finland	11.76%	8.17%	2.81%	2.26%	4.04%	
Ireland	12.16%	5.15%	5.04%	4.75%	10.88%	
Japan	1.79%	3.17%	1.99%	0.47%	-1.08%	
Netherlands	-1.54%	1.93%	4.03%	5.14%	5.14%	
Sweden	6.94%	8.06%	3.19%	3.79%	3.36%	
United States	7.94%	3.67%	3.63%	2.53%	2.78%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 6: PIS1 Value Added at Basic Prices, Percentages of the GDP

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Notes
1	Supply of ICT Devices, Systems and Networks					
<i>Average</i>	4.08%	4.54%	4.80%	6.51%	6.83%	
Austria	3.73%	4.20%	4.35%	4.74%	5.03%	a
Denmark	2.86%	3.57%	3.84%	4.62%	5.06%	a
Finland	3.34%	3.96%	4.50%	8.22%	10.38%	a
Ireland	4.66%	5.72%	5.70%	9.60%	8.80%	
Japan	5.84%	6.20%	6.46%	7.09%	7.07%	
Netherlands	3.33%	3.59%	3.68%	4.76%	5.22%	a
Sweden	3.80%	3.81%	4.48%	6.74%	6.81%	
United States	5.05%	5.26%	5.35%	6.33%	6.28%	
1b	New ICT Manufacturing Industries					
<i>Average</i>	1.47%	1.72%	1.69%	2.62%	2.36%	
Austria	1.15%	1.27%	1.23%	1.48%	1.41%	a
Denmark	0.49%	0.51%	0.57%	1.06%	1.05%	a
Finland	0.67%	0.98%	1.28%	4.16%	5.03%	a
Ireland	2.32%	3.49%	3.36%	5.67%	4.52%	
Japan	2.58%	3.06%	2.97%	3.00%	2.62%	
Netherlands	1.19%	1.12%	0.88%	0.99%	0.66%	a
Sweden	1.10%	1.08%	1.10%	2.31%	1.73%	
United States	2.26%	2.27%	2.14%	2.32%	1.88%	
1c&d	ICT Software, DP, Sel. Mail/Courier etc. and connectivity services					
<i>Average</i>	2.61%	2.81%	3.10%	3.89%	4.47%	
Austria	2.59%	2.93%	3.13%	3.26%	3.62%	
Denmark	2.37%	3.06%	3.27%	3.57%	4.01%	
Finland	2.67%	2.98%	3.22%	4.07%	5.35%	
Ireland	2.34%	2.23%	2.33%	3.93%	4.28%	
Japan	3.26%	3.14%	3.49%	4.09%	4.46%	
Netherlands	2.14%	2.46%	2.80%	3.78%	4.57%	
Sweden	2.70%	2.73%	3.38%	4.43%	5.08%	
United States	2.78%	2.99%	3.21%	4.00%	4.40%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 7: PIS2 Value Added at Basic Prices, Percentages of the GDP

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Notes
2	Core PIS information Industries					
<i>Average</i>	12.47%	13.63%	14.76%	15.38%	15.94%	
Austria	11.63%	12.46%	13.06%	13.91%	14.93%	
Denmark	14.01%	14.73%	15.70%	16.08%	16.49%	
Finland	11.43%	12.91%	13.96%	14.14%	14.23%	
Ireland	14.00%	15.11%	16.21%	15.53%	15.43%	h
Japan	10.30%	10.78%	11.48%	11.47%	11.93%	f
Netherlands	11.91%	14.93%	16.55%	18.19%	18.26%	
Sweden	12.96%	12.79%	14.51%	16.14%	17.63%	
United States	13.51%	15.32%	16.62%	17.61%	18.61%	f
2a	Consumer Media and Cultural Industries					
<i>Average</i>	2.49%	2.85%	2.96%	3.17%	3.05%	
Austria	1.85%	2.14%	2.28%	2.39%	2.48%	
Denmark	3.14%	3.25%	3.45%	3.41%	3.10%	
Finland	3.43%	3.70%	3.62%	3.42%	3.20%	
Ireland	2.64%	3.00%	3.36%	5.04%	4.67%	
Japan	3.04%	3.48%	3.35%	3.06%	3.03%	
Netherlands	1.45%	2.65%	2.92%	2.90%	2.75%	
Sweden	2.20%	2.25%	2.32%	2.64%	2.63%	
United States	2.15%	2.31%	2.39%	2.53%	2.56%	
2b	Instrumental/ Producer IC Services					
<i>Average</i>	4.85%	5.70%	6.54%	7.31%	7.95%	
Austria	3.63%	4.17%	5.03%	6.03%	7.14%	
Denmark	5.06%	5.97%	6.66%	7.08%	7.71%	
Finland	3.41%	4.25%	4.74%	5.70%	6.20%	
Ireland	7.25%	7.63%	7.47%	6.27%	6.79%	h
Japan	3.33%	3.27%	4.05%	4.71%	5.11%	f
Netherlands	5.11%	7.67%	9.33%	11.20%	11.18%	
Sweden	4.94%	5.19%	6.72%	8.20%	9.40%	
United States	6.03%	7.46%	8.34%	9.31%	10.08%	f
2c	Mixed' Producer-Consumer IC Services					
<i>Average</i>	5.14%	5.08%	5.26%	4.90%	4.94%	
Austria	6.15%	6.15%	5.75%	5.50%	5.32%	
Denmark	5.80%	5.51%	5.58%	5.59%	5.68%	
Finland	4.60%	4.96%	5.60%	5.01%	4.84%	
Ireland	4.11%	4.48%	5.38%	4.22%	3.97%	
Japan	3.93%	4.03%	4.08%	3.70%	3.78%	
Netherlands	5.35%	4.60%	4.31%	4.09%	4.33%	
Sweden	5.82%	5.35%	5.47%	5.30%	5.61%	
United States	5.32%	5.56%	5.90%	5.77%	5.98%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 8: PIS1 Relative Employment (Percentage of the total in the economy)

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Notes
1	Supply of ICT Devices, Systems and Networks					
<i>Average</i>	4.29%	4.26%	5.04%	5.04%	5.13%	
Austria	4.21%	3.97%	3.93%	3.93%	4.03%	
Denmark	3.51%	3.59%	4.23%	4.23%	4.54%	a
Finland	3.91%	4.40%	5.89%	5.89%	6.40%	a
Ireland	5.43%	5.36%	7.08%	7.08%	6.75%	
Japan	5.96%	5.75%	5.65%	5.65%	5.74%	
Netherlands	3.53%	3.38%	4.28%	4.28%	4.36%	a
Sweden	4.24%	4.24%	5.43%	5.43%	5.50%	a
United States	3.53%	3.37%	3.81%	3.81%	3.70%	
1b	New ICT Manufacturing Industries					
<i>Average</i>	1.64%	1.63%	1.61%	2.00%	1.74%	
Austria	1.74%	1.63%	1.50%	1.32%	1.23%	
Denmark	0.61%	0.59%	0.66%	1.08%	0.96%	a
Finland	0.62%	0.76%	1.08%	2.27%	2.29%	a
Ireland	2.61%	2.50%	2.76%	3.82%	3.25%	
Japan	3.13%	3.43%	3.14%	2.86%	2.45%	
Netherlands	1.15%	1.09%	0.98%	1.27%	1.05%	a
Sweden	1.27%	1.26%	1.23%	1.93%	1.52%	a
United States	2.00%	1.81%	1.53%	1.44%	1.17%	
1c&d	ICT Software, DP, Sel. Mail/Courier etc. and connectivity services					
<i>Average</i>	2.56%	2.66%	2.65%	3.04%	3.39%	
Austria	2.51%	2.59%	2.47%	2.61%	2.79%	
Denmark	2.73%	2.92%	2.93%	3.15%	3.58%	
Finland	2.88%	3.15%	3.32%	3.62%	4.11%	
Ireland	3.15%	2.92%	2.61%	3.26%	3.50%	
Japan	2.27%	2.52%	2.60%	2.80%	3.30%	
Netherlands	2.43%	2.43%	2.40%	3.01%	3.31%	
Sweden	2.79%	2.98%	3.01%	3.50%	3.98%	
United States	1.77%	1.72%	1.83%	2.36%	2.53%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 9: PIS2 Relative Employment (Percentage of the total in the economy)

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Notes
2	Core PIS information Industries					
<i>Average</i>	15.86%	17.11%	18.35%	19.76%	20.81%	
Austria	14.09%	14.90%	14.77%	15.52%	17.17%	
Denmark	16.68%	16.97%	18.13%	19.20%	20.57%	
Finland	13.54%	15.11%	17.18%	18.41%	19.41%	
Ireland	15.76%	16.96%	18.15%	19.07%	19.22%	
Japan	12.71%	13.28%	13.96%	14.86%	16.06%	f
Netherlands	15.86%	19.03%	20.42%	24.87%	24.73%	i
Sweden	19.80%	20.69%	22.23%	22.50%	24.22%	
United States	18.45%	19.96%	21.96%	23.70%	25.12%	f
2a	Consumer Media and Cultural Industries					
<i>Average</i>	2.88%	2.96%	3.16%	3.37%	3.22%	
Austria	2.30%	2.40%	2.36%	2.36%	2.27%	
Denmark	3.47%	3.47%	3.53%	3.54%	3.39%	
Finland	3.71%	4.03%	4.23%	4.01%	3.92%	
Ireland	2.85%	2.92%	3.42%	4.29%	4.11%	
Japan	2.89%	3.18%	3.52%	3.56%	3.28%	
Netherlands	1.92%	1.84%	2.15%	3.11%	2.91%	i
Sweden	3.37%	3.22%	3.31%	3.21%	3.17%	
United States	2.56%	2.59%	2.73%	2.86%	2.69%	
2b	Instrumental/ Producer IC Services					
<i>Average</i>	5.77%	7.06%	8.01%	9.37%	10.40%	
Austria	4.76%	5.45%	6.00%	6.97%	8.51%	
Denmark	5.55%	6.08%	6.94%	7.92%	9.39%	
Finland	3.96%	4.93%	5.73%	6.77%	7.91%	
Ireland	5.00%	6.18%	6.77%	7.02%	7.74%	
Japan	5.56%	6.07%	6.67%	7.62%	9.02%	f
Netherlands	7.14%	10.89%	12.45%	16.40%	16.22%	
Sweden	6.22%	7.37%	8.65%	9.84%	11.04%	
United States	8.00%	9.49%	10.88%	12.46%	13.39%	f
2c	Mixed' Producer-Consumer IC Services					
<i>Average</i>	7.21%	7.10%	7.18%	7.02%	7.19%	
Austria	7.04%	7.05%	6.42%	6.18%	6.39%	
Denmark	7.66%	7.42%	7.66%	7.73%	7.79%	
Finland	5.87%	6.15%	7.21%	7.62%	7.58%	
Ireland	7.91%	7.86%	7.96%	7.76%	7.36%	
Japan	4.26%	4.03%	3.76%	3.68%	3.76%	
Netherlands	6.80%	6.30%	5.82%	5.35%	5.60%	
Sweden	10.21%	10.10%	10.27%	9.45%	10.01%	
United States	7.90%	7.88%	8.35%	8.38%	9.05%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 10: PIS1 Share of Value Added, Percentage Change

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Average 1981-05	Notes
1	Supply of ICT Devices, Systems and Networks						
<i>Average</i>	0.63%	0.28%	0.82%	1.93%	-0.75%	0.58%	
Austria	0.45%	0.34%	0.35%	0.25%	-0.11%	0.25%	a
Denmark	0.65%	0.69%	0.17%	0.64%	0.20%	0.47%	a
Finland	0.82%	0.41%	1.42%	5.13%	-0.80%	1.40%	a
Ireland	1.71%	-0.07%	2.16%	4.85%	-4.42%	0.85%	
Japan	0.36%	0.61%	0.11%	0.75%	-0.27%	0.31%	
Netherlands	0.06%	0.31%	0.33%	1.41%	-0.21%	0.38%	a
Sweden	0.07%	0.13%	1.56%	1.33%	0.36%	0.69%	
United States	0.91%	-0.17%	0.49%	1.11%	-0.77%	0.31%	
1b	New ICT Manufacturing Industries						
<i>Average</i>	0.44%	0.06%	0.53%	0.77%	-0.80%	0.20%	
Austria	0.15%	0.11%	0.10%	0.22%	-0.36%	0.04%	a
Denmark	-0.02%	0.02%	0.44%	0.05%	-0.11%	0.08%	a
Finland	0.41%	0.26%	1.26%	3.49%	-1.11%	0.86%	a
Ireland	1.89%	0.22%	1.81%	1.88%	-3.25%	0.51%	
Japan	0.67%	0.53%	-0.45%	0.25%	-0.45%	0.11%	
Netherlands	-0.03%	-0.21%	0.07%	0.10%	-0.60%	-0.13%	a
Sweden	0.11%	-0.16%	0.90%	0.01%	0.16%	0.20%	
United States	0.33%	-0.25%	0.14%	0.14%	-0.70%	-0.07%	
1c&d	ICT Software, DP, Sel. Mail/Courier etc. and connectivity services						
<i>Average</i>	0.19%	0.22%	0.29%	1.17%	0.05%	0.38%	
Austria	0.30%	0.23%	0.25%	0.03%	0.24%	0.21%	
Denmark	0.67%	0.66%	-0.27%	0.59%	0.31%	0.39%	
Finland	0.41%	0.15%	0.16%	1.64%	0.31%	0.53%	
Ireland	-0.18%	-0.29%	0.36%	2.97%	-1.17%	0.34%	
Japan	-0.32%	0.08%	0.56%	0.50%	0.18%	0.20%	
Netherlands	0.09%	0.52%	0.25%	1.31%	0.39%	0.51%	
Sweden	-0.03%	0.29%	0.66%	1.32%	0.20%	0.49%	
United States	0.58%	0.08%	0.35%	0.97%	-0.07%	0.38%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 11: PIS2 Share of Value Added, Percentage Change

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	SUM 1981 05	Notes
2	Core PIS information Industries						
<i>Average</i>	0.58%	1.45%	0.64%	0.86%	0.42%	0.79%	
Austria	0.65%	0.79%	0.68%	1.28%	0.37%	0.75%	
Denmark	-0.24%	1.69%	0.16%	0.79%	0.19%	0.52%	
Finland	1.23%	1.38%	0.58%	-0.18%	0.99%	0.80%	
Ireland	0.49%	1.18%	0.34%	-0.84%	1.15%	0.46%	h
Japan	1.34%	0.59%	-0.29%	0.85%	0.15%	0.53%	f
Netherlands	-1.16%	3.99%	1.29%	1.29%	-0.08%	1.07%	
Sweden	0.13%	0.31%	1.68%	2.38%	0.75%	1.05%	
United States	2.25%	1.64%	0.72%	1.33%	-0.13%	1.16%	f
2a	Consumer Media and Cultural Industries						
<i>Average</i>	0.09%	0.45%	0.01%	0.25%	-0.21%	0.12%	
Austria	-0.08%	0.45%	0.04%	0.21%	0.01%	0.13%	
Denmark	-0.16%	0.36%	-0.02%	-0.10%	-0.32%	-0.05%	
Finland	0.29%	0.16%	-0.13%	-0.38%	-0.03%	-0.02%	
Ireland	0.09%	0.37%	0.46%	2.01%	-0.86%	0.41%	
Japan	0.36%	0.65%	-0.65%	-0.09%	0.07%	0.07%	
Netherlands	-0.15%	1.53%	-0.06%	-0.02%	-0.18%	0.22%	
Sweden	0.07%	-0.02%	0.35%	0.18%	-0.10%	0.09%	
United States	0.27%	0.10%	0.06%	0.23%	-0.24%	0.08%	
2b	Instrumental/ Producer IC Services						
<i>Average</i>	0.59%	0.92%	0.64%	0.96%	0.44%	0.71%	
Austria	0.34%	0.76%	0.87%	1.27%	0.52%	0.75%	
Denmark	0.32%	1.37%	0.02%	0.89%	0.42%	0.60%	
Finland	0.58%	0.94%	0.41%	0.84%	0.74%	0.70%	
Ireland	0.39%	-0.21%	-0.49%	-1.12%	1.34%	-0.02%	h
Japan	1.03%	-0.26%	0.75%	0.90%	0.05%	0.49%	f
Netherlands	0.05%	3.02%	1.38%	1.53%	-0.23%	1.15%	
Sweden	0.47%	0.39%	1.69%	2.16%	0.44%	1.03%	
United States	1.58%	1.37%	0.50%	1.19%	0.21%	0.97%	f
2c	Mixed' Producer-Consumer IC Services						
<i>Average</i>	-0.22%	-0.02%	0.04%	-0.38%	0.23%	-0.07%	
Austria	0.39%	-0.42%	-0.23%	-0.21%	-0.16%	-0.12%	
Denmark	-0.41%	-0.03%	0.15%	-0.01%	0.09%	-0.04%	
Finland	0.36%	0.27%	0.30%	-0.64%	0.29%	0.12%	
Ireland	0.00%	1.01%	0.36%	-1.73%	0.67%	0.06%	
Japan	-1.06%	-0.55%	-0.03%	-0.23%	0.34%	-0.31%	
Netherlands	-1.06%	-0.55%	-0.03%	-0.23%	0.34%	-0.31%	
Sweden	-0.41%	-0.07%	-0.35%	0.04%	0.41%	-0.08%	
United States	0.40%	0.17%	0.16%	-0.08%	-0.10%	0.11%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 12: PIS1 Share of Employment, Percentage Change

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	Average 1981-05	Notes
1	Supply of ICT Devices, Systems and Networks						
<i>Average</i>	0.20%	-0.02%	0.26%	0.90%	-0.49%	0.17%	
Austria	-0.08%	-0.03%	-0.39%	0.39%	-0.29%	-0.08%	
Denmark	0.03%	0.22%	0.34%	0.71%	-0.14%	0.23%	a
Finland	0.38%	0.19%	1.35%	1.22%	-0.24%	0.58%	a
Ireland	-0.24%	-0.43%	0.46%	1.89%	-1.29%	0.08%	
Japan	1.05%	0.22%	-0.61%	0.36%	-0.07%	0.19%	
Netherlands	0.15%	-0.17%	0.38%	0.83%	-0.49%	0.14%	a
Sweden	0.13%	0.09%	0.62%	1.07%	-0.80%	0.22%	a
United States	0.15%	-0.28%	-0.07%	0.71%	-0.57%	-0.01%	
1b	New ICT Manufacturing Industries						
<i>Average</i>	0.03%	-0.07%	0.31%	0.11%	-0.40%	0.00%	
Austria	-0.23%	-0.01%	-0.28%	-0.02%	-0.15%	-0.14%	
Denmark	-0.02%	-0.05%	0.54%	-0.04%	-0.18%	0.05%	a
Finland	0.06%	0.06%	1.16%	0.62%	-0.32%	0.32%	a
Ireland	-0.36%	-0.01%	0.75%	0.75%	-1.06%	0.01%	
Japan	0.89%	-0.11%	-0.49%	-0.13%	-0.47%	-0.06%	
Netherlands	-0.11%	-0.08%	0.36%	-0.23%	-0.21%	-0.05%	a
Sweden	-0.04%	-0.10%	0.66%	0.00%	-0.53%	0.00%	a
United States	0.08%	-0.29%	-0.24%	-0.09%	-0.30%	-0.17%	
1c&d	ICT Software, DP, Sel. Mail/Courier etc. and connectivity services						
<i>Average</i>	0.16%	0.05%	-0.05%	0.79%	-0.09%	0.17%	
Austria	0.15%	-0.03%	-0.11%	0.41%	-0.14%	0.06%	
Denmark	0.06%	0.28%	-0.20%	0.75%	0.04%	0.18%	
Finland	0.31%	0.13%	0.19%	0.60%	0.08%	0.26%	
Ireland	0.12%	-0.41%	-0.30%	1.14%	-0.23%	0.07%	
Japan	0.16%	0.33%	-0.12%	0.49%	0.40%	0.25%	
Netherlands	0.25%	-0.09%	0.02%	1.06%	-0.28%	0.19%	
Sweden	0.17%	0.19%	-0.04%	1.06%	-0.27%	0.22%	
United States	0.07%	0.01%	0.17%	0.79%	-0.27%	0.15%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Table 13: PIS2 Share of Employment, Percentage Change

PIS Sector	1981-85	1986-90	1991-95	1996-00	2001-05	SUM 1981-05	Notes
2	Core PIS information Industries						
<i>Average</i>	1.10%	1.28%	1.50%	1.12%	1.04%	1.21%	
Austria	1.21%	0.10%	-0.02%	1.63%	1.41%	0.87%	
Denmark	-0.03%	0.99%	1.10%	1.54%	1.05%	0.93%	
Finland	1.34%	1.77%	2.02%	0.91%	0.97%	1.40%	
Ireland	0.88%	1.89%	1.37%	0.25%	0.30%	0.94%	
Japan	0.76%	0.55%	0.77%	0.87%	1.46%	0.88%	f
Netherlands	2.60%	1.73%	3.91%	1.50%	0.10%	1.97%	i
Sweden	1.00%	1.12%	1.10%	0.56%	1.94%	1.14%	
United States	1.07%	2.05%	1.77%	1.68%	1.09%	1.53%	f
2a	Consumer Media and Cultural Industries						
<i>Average</i>	0.02%	0.11%	0.38%	-0.05%	-0.17%	0.06%	
Austria	0.04%	0.06%	-0.04%	0.06%	-0.23%	-0.02%	
Denmark	-0.18%	0.09%	-0.02%	0.09%	-0.28%	-0.06%	
Finland	0.40%	0.20%	0.07%	-0.14%	-0.15%	0.07%	
Ireland	-0.20%	0.26%	0.97%	0.26%	-0.06%	0.25%	
Japan	0.29%	0.29%	0.23%	-0.08%	-0.31%	0.09%	
Netherlands	-0.13%	0.02%	1.46%	-0.26%	-0.19%	0.18%	i
Sweden	-0.12%	-0.12%	0.20%	-0.35%	0.09%	-0.06%	
United States	0.04%	0.10%	0.18%	0.02%	-0.21%	0.03%	
2b	Instrumental/ Producer IC Services						
<i>Average</i>	0.96%	1.29%	0.93%	1.43%	0.83%	1.09%	
Austria	0.91%	0.43%	0.52%	1.64%	1.31%	0.96%	
Denmark	0.25%	0.89%	0.71%	1.40%	1.29%	0.91%	
Finland	0.77%	1.22%	0.58%	1.19%	1.06%	0.96%	
Ireland	0.94%	1.64%	-0.38%	1.10%	0.20%	0.70%	
Japan	0.63%	0.52%	0.72%	0.96%	1.63%	0.90%	f
Netherlands	2.65%	2.33%	2.83%	2.13%	-0.17%	1.95%	
Sweden	0.48%	1.50%	1.22%	1.47%	0.71%	1.07%	
United States	1.09%	1.79%	1.22%	1.57%	0.59%	1.25%	f
2c	Mixed' Producer-Consumer IC Services						
<i>Average</i>	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	
Austria	0.99%	0.99%	0.99%	0.99%	0.99%	0.99%	
Denmark	1.40%	1.40%	1.40%	1.40%	1.40%	1.40%	
Finland	-0.32%	-0.32%	-0.32%	-0.32%	-0.32%	-0.32%	
Ireland	0.18%	0.18%	0.18%	0.18%	0.18%	0.18%	
Japan	1.09%	1.09%	1.09%	1.09%	1.09%	1.09%	
Netherlands	1.09%	1.09%	1.09%	1.09%	1.09%	1.09%	
Sweden	1.10%	1.10%	1.10%	1.10%	1.10%	1.10%	
United States	0.87%	0.87%	0.87%	0.87%	0.87%	0.87%	

Source: own calculations based on data from EU KLEMS database, 2008, notes on page 34.

Notes tables 2 to 13

- a) The averages for the intervals 1981-1985, 1986-1990, 1991-1995 do not include the output of the 'Scientific instruments' (331t3) industries.
- b) This category includes data about the output of the 'Recreational, cultural and sporting activities' (92) industries, but not the output of the 'Print., publishing and recorded media' industries.
- c) The series does not include data for the 'Recreational, cultural and sporting activities' for the period from 1981 to 1987 included.
- d) Data about the output of the 'Research and Development' and of the 'Other Business Activities' industries, between 1980 and 1995, in the EU KLEMS database are estimates.
- e) Data about the output of the 'Activities auxiliary to financial intermediation' is not included.
- f) The series do not include data about the output of the 'Activities auxiliary to financial intermediation' and 'Activities of member organisations nec' industries for the period from 1981 to 1987 included.
- g) The data about the price trends for the years 1981 and 1982 for the 'Office, accounting & computing machinery' is not included in the average as it is missing from the database.
- h) The averages for the intervals 1981-1985, 1986-1990, 1991-1995 do not include data about the output trends of the 'Research and Development' industries.
- i) The series does not include data about the employment in the 'Recreational, cultural and sporting activities' for the period from 1981 to 1994.

Table 14: Total EPO patents per thousands of employees.

	1985	1990	1996	2000	2005	2006
Austria	6.22	12.77	11.99	7.12	17.96	19.38
Denmark	2.83	4.19	8.41	7.30	16.64	17.25
Finland	1.97	5.77	14.43	11.66	27.95	30.19
Ireland	0.90	2.17	3.02	1.95	4.64	4.34
Netherlands	10.20	14.40	13.89	9.51	16.68	18.79
Sweden	8.25	12.66	16.17	13.11	27.93	28.92
United States	3.13	4.73	7.71	5.06	8.81	9.44

Sources: calculated by the authors. Data on patents is taken from NSF Science & Eng Indicators, 2008, Vol. 2, Appendix; data on the number of employees: for the years 1985, 1990 and for all countries; and for the years 1996-2006 for the US, the data is extracted from the KLEMS database; 1996, 200, 2005 and 2006 for AT, DK, FI, IR, NE, SW from the Eurostat, Labour Force Survey database.

Table 15: Biotechnology-related patents (EPO & USPTO) per million of employees

	1985	1990	1996	2000	2005	2006
Austria	1.06	3.66	7.41	8.89	12.03	8.66
Denmark	3.41	4.54	13.35	37.96	32.34	41.35
Finland	2.42	3.73	15.43	8.99	12.50	14.32
Ireland	0.00	1.10	4.53	2.95	5.12	8.83
Netherlands	2.84	5.65	7.81	14.87	13.19	14.53
Sweden	4.58	4.86	9.40	14.62	16.56	18.96
United States	4.33	7.11	20.07	29.67	24.70	30.47

Sources: estimated by the authors; see table 15.

Table 16: ICT-related Patents (EPO and USPTO) per million of employees

	1985	1990	1996	2000	2005	2006
Austria	13.11	20.62	n.a.	33.42	60.40	67.46
Denmark	10.22	15.27	n.a.	29.49	69.76	80.92
Finland	10.64	21.42	n.a.	154.95	320.73	378.56
Ireland	9.52	18.65	n.a.	27.18	49.18	59.35
Netherlands	70.24	102.12	n.a.	80.30	107.88	140.66
Sweden	39.57	36.53	n.a.	146.22	184.74	201.38
United States	79.24	97.24	n.a.	220.31	260.41	308.60

Sources: estimated by the authors; see table 15

Table 17: Imports, Exports and Balance of Payments (in percentage of the GDP) in Telecommunication Services

Telecommunication Services					
Averages		1991 to 1995	1996 to 2000	2001 to 2005	2006
Austria	Net	n.a.	n.a.	0.09%	0.03%
	Exports	n.a.	n.a.	0.24%	0.30%
	Imports	n.a.	n.a.	0.15%	0.27%
Denmark	Net	n.a.	n.a.	-0.03%	-0.01%
	Exports	n.a.	n.a.	0.24%	0.23%
	Imports	n.a.	n.a.	0.27%	0.24%
Finland	Net	n.a.	-0.05%	-0.06%	n.a.
	Exports	n.a.	0.11%	0.13%	n.a.
	Imports	n.a.	0.16%	0.19%	n.a.
Ireland	Net	n.a.	-0.18%	0.09%	-0.20%
	Exports	n.a.	0.21%	0.54%	0.24%
	Imports	n.a.	0.39%	0.45%	0.43%
Netherlands	Net	n.a.	n.a.	0.02%	0.04%
	Exports	n.a.	n.a.	0.29%	0.30%
	Imports	n.a.	n.a.	0.28%	0.26%
Sweden	Net	n.a.	-0.08%	-0.03%	-0.06%
	Exports	n.a.	0.19%	0.27%	0.36%
	Imports	n.a.	0.27%	0.30%	0.42%
United States	Net	-0.05%	-0.04%	0.00%	0.01%
	Exports	0.05%	0.05%	0.04%	0.05%
	Imports	0.10%	0.08%	0.04%	0.03%

Source: the authors, based on data published in OECD Statistics on International Trade in Services, 1970 - 2006, 2008 Edition.

Table 18: Imports, Exports and Balance of Payments (in percentage of the GDP) in Financial Services

Financial Services					
Averages		1991 to 1995	1996 to 2000	2001 to 2005	2006
Austria	Net	-0.06%	0.03%	0.08%	0.06%
	Exports	0.62%	0.43%	0.44%	0.25%
	Imports	0.68%	0.40%	0.36%	0.19%
Denmark	Net	n.a.	n.a.	-0.06%	-0.05%
	Exports	n.a.	n.a.	0.18%	0.08%
	Imports	n.a.	n.a.	0.24%	0.12%
Finland	Net	-0.01%	-0.01%	0.01%	-0.05%
	Exports	0.29%	0.04%	0.05%	0.04%
	Imports	0.30%	0.04%	0.04%	0.08%
Ireland	Net	-0.53%	-0.08%	1.10%	1.38%
	Exports	0.34%	0.94%	2.48%	3.49%
	Imports	0.88%	1.02%	1.38%	2.12%
Netherlands	Net	0.00%	-0.02%	-0.07%	-0.07%
	Exports	0.09%	0.14%	0.17%	0.21%
	Imports	0.08%	0.17%	0.24%	0.28%
Sweden	Net	0.00%	0.03%	0.09%	0.20%
	Exports	0.09%	0.19%	0.30%	0.46%
	Imports	0.09%	0.16%	0.21%	0.26%
United States	Net	0.05%	0.08%	0.14%	0.22%
	Exports	0.08%	0.16%	0.24%	0.33%
	Imports	0.03%	0.09%	0.10%	0.11%

Source: the authors, based on data published in OECD Statistics on International Trade in Services, 1970 - 2006, 2008 Edition.

Table 19: Imports, Exports and Balance of Payments (in percentage of the GDP) in Computer and Information Services

Computer and Information Services					
Averages		1991 to 1995	1996 to 2000	2001 to 2005	2006
Austria	Net	-0.03%	-0.05%	-0.06%	0.14%
	Exports	0.03%	0.05%	0.09%	0.47%
	Imports	0.06%	0.10%	0.15%	0.33%
Denmark	Net	n.a.	n.a.	0.03%	-0.10%
	Exports	n.a.	n.a.	0.43%	0.44%
	Imports	n.a.	n.a.	0.40%	0.54%
Finland	Net	-0.04%	0.24%	0.06%	0.17%
	Exports	0.50%	0.69%	0.43%	0.70%
	Imports	0.54%	0.45%	0.36%	0.53%
Ireland	Net	-0.18%	1.89%	8.99%	9.15%
	Exports	0.08%	2.30%	9.31%	9.45%
	Imports	0.26%	0.41%	0.32%	0.30%
Netherlands	Net	0.02%	0.02%	0.01%	0.02%
	Exports	0.15%	0.24%	0.46%	0.58%
	Imports	0.13%	0.22%	0.45%	0.55%
Sweden	Net	0.02%	0.02%	0.28%	0.34%
	Exports	0.08%	0.29%	0.66%	0.91%
	Imports	0.07%	0.27%	0.39%	0.57%
United States	Net	0.03%	0.03%	0.02%	-0.01%
	Exports	0.03%	0.06%	0.07%	0.08%
	Imports	0.00%	0.03%	0.05%	0.08%

Source: the authors, based on data published in OECD Statistics on International Trade in Services, 1970 - 2006, 2008 Edition.

Table 20: Imports, Exports and Balance of Payments (in percentage of the GDP) in Personal, Cultural and Recreational Services

Personal, Cultural and Recreational Services					
Averages		1991 to 1995	1996 to 2000	2001 to 2005	2006
Austria	Net	-0.01%	0.00%	-0.01%	-0.15%
	Exports	0.07%	0.10%	0.11%	0.08%
	Imports	0.07%	0.10%	0.12%	0.23%
Denmark	Net	n.a.	n.a.	-0.11%	-0.30%
	Exports	n.a.	n.a.	0.09%	0.14%
	Imports	n.a.	n.a.	0.20%	0.44%
Finland	Net	-0.01%	-0.07%	-0.01%	0.00%
	Exports	0.07%	0.04%	0.01%	0.01%
	Imports	0.08%	0.11%	0.02%	0.01%
Ireland	Net	-0.30%	-0.35%	0.17%	0.04%
	Exports	0.06%	0.09%	0.22%	0.10%
	Imports	0.36%	0.44%	0.05%	0.06%
Netherlands	Net	-0.01%	-0.01%	-0.02%	-0.01%
	Exports	0.11%	0.13%	0.13%	0.13%
	Imports	0.12%	0.13%	0.14%	0.15%
Sweden	Net	0.00%	-0.01%	0.02%	0.03%
	Exports	0.05%	0.04%	0.05%	0.07%
	Imports	0.05%	0.06%	0.04%	0.04%
United States	Net	0.05%	0.08%	0.09%	0.08%
	Exports	0.05%	0.08%	0.09%	0.09%
	Imports	0.00%	0.00%	0.00%	0.01%

Source: the authors, based on data published in OECD Statistics on International Trade in Services, 1970 - 2006, 2008 Edition.

Table 21: Imports, Exports and Balance of Payments (in percentage of the GDP) in Royalties and Licence Fees

Royalties and Licence Fees					
Averages		1991 to 1995	1996 to 2000	2001 to 2005	2006
Austria	Net	-0.16%	-0.25%	-0.38%	-0.24%
	Exports	0.06%	0.07%	0.06%	0.17%
	Imports	0.22%	0.32%	0.44%	0.41%
Denmark	Net	n.a.	n.a.	0.06%	0.13%
	Exports	n.a.	n.a.	0.43%	0.61%
	Imports	n.a.	n.a.	0.38%	0.48%
Finland	Net	-0.24%	-0.08%	0.00%	-0.11%
	Exports	0.07%	0.29%	0.45%	0.51%
	Imports	0.31%	0.37%	0.45%	0.62%
Ireland	Net	-2.95%	-6.13%	-9.41%	-8.88%
	Exports	0.14%	0.29%	0.23%	0.46%
	Imports	3.09%	6.42%	9.64%	9.34%
Netherlands	Net	-0.18%	-0.14%	0.09%	0.34%
	Exports	0.56%	0.59%	1.19%	1.62%
	Imports	0.74%	0.73%	1.10%	1.27%
Sweden	Net	0.05%	0.07%	0.40%	0.59%
	Exports	0.36%	0.43%	0.79%	1.01%
	Imports	0.31%	0.36%	0.39%	0.43%
United States	Net	0.27%	0.29%	0.26%	0.27%
	Exports	0.35%	0.42%	0.44%	0.48%
	Imports	0.08%	0.13%	0.18%	0.20%

Source: the authors, based on data published in OECD Statistics on International Trade in Services, 1970 - 2006, 2008 Edition.

Table 22: Imports, Exports and Balance of Payments (in percentage of the GDP) in Research and Development

Research and Development					
Averages		1991 to 1995	1996 to 2000	2001 to 2005	2006
Austria	Net	n.a.	n.a.	0.10%	0.43%
	Exports	n.a.	n.a.	0.35%	0.53%
	Imports	n.a.	n.a.	0.25%	0.10%
Denmark	Net	n.a.	n.a.	0.18%	0.15%
	Exports	n.a.	n.a.	0.40%	0.39%
	Imports	n.a.	n.a.	0.22%	0.24%
Finland	Net	-0.04%	-0.04%	-0.03%	n.a.
	Exports	0.05%	0.06%	0.17%	n.a.
	Imports	0.09%	0.11%	0.20%	n.a.
Ireland	Net	-0.12%	-0.27%	-1.85%	-1.98%
	Exports	0.11%	0.16%	0.27%	0.13%
	Imports	0.24%	0.43%	2.09%	2.11%
Netherlands	Net	0.12%	0.10%	-0.07%	0.11%
	Exports	0.31%	0.33%	0.64%	0.72%
	Imports	0.19%	0.23%	0.71%	0.61%
Sweden	Net	0.01%	-0.01%	-0.23%	-0.42%
	Exports	0.06%	0.20%	0.63%	0.68%
	Imports	0.05%	0.21%	0.86%	1.10%
United States	Net	0.00%	0.00%	0.04%	0.03%
	Exports	0.01%	0.01%	0.08%	0.10%
	Imports	0.00%	0.01%	0.04%	0.07%

Source: the authors, based on data published in OECD Statistics on International Trade in Services, 1970 - 2006, 2008 Edition.

Table 23: Imports, Exports and Balance of Payments (in percentage of the GDP) in Other Business Services

Other Business Services					
Averages		1991 to 1995	1996 to 2000	2001 to 2005	2006
Austria	Net	1.27%	0.54%	0.79%	1.81%
	Exports	4.11%	2.31%	2.76%	3.92%
	Imports	2.83%	1.77%	1.97%	2.11%
Denmark	Net	n.a.	0.40%	0.26%	0.07%
	Exports	n.a.	1.91%	2.30%	2.97%
	Imports	n.a.	1.51%	2.04%	2.91%
Finland	Net	-0.71%	-0.61%	1.57%	2.34%
	Exports	1.00%	1.18%	3.28%	4.19%
	Imports	1.71%	1.80%	1.71%	1.85%
Ireland	Net	-2.52%	-6.09%	-9.96%	-6.29%
	Exports	1.28%	2.06%	5.40%	8.40%
	Imports	3.80%	8.15%	15.36%	14.69%
Netherlands	Net	0.30%	0.17%	0.00%	0.19%
	Exports	3.52%	3.66%	4.83%	5.34%
	Imports	3.22%	3.49%	4.84%	5.15%
Sweden	Net	0.00%	-0.13%	0.11%	1.11%
	Exports	0.79%	1.96%	3.73%	4.60%
	Imports	0.79%	2.09%	3.61%	3.48%
United States	Net	0.13%	0.18%	0.21%	0.25%
	Exports	0.35%	0.42%	0.49%	0.61%
	Imports	0.22%	0.24%	0.28%	0.36%
includes: Merchanting and other trade-related services; Operational leasing services; Miscellaneous business, professional and technical services; Legal, accounting, management consulting and public relations services; Business and management consultancy and public relations services; Advertising, market research and public opinion polling; Research and development; Architectural, engineering and other technical services					

Source: the authors, based on data published in OECD Statistics on International Trade in Services, 1970 - 2006, 2008 Edition.